What is Run Test in Statistics – A Simple Explanation with Step by Step Examples

Run test is a statistical test used to determine of the data obtained from a sample is random. That is why it is called Run Test for Randomness. Randomness of the data is determined based on the number and nature of runs present in the data of interest.

What is a Run?

A run is a sequence of similar or like events, items or symbols that is preceded by and followed by an event, item or symbol of a different type, or by none at all. Randomness of the series is unlikely when there appear to be either too many or two few runs. In this case, a run test needs to be carried out to determine the randomness.

The Run Test when performed helps us to decide whether a sequence of events, items or symbol is the result of a random process.

Example of Runs

A data scientist carrying out a research interviewed 10 persons during a survey. We denote the genders of the people by M for men and W for women.

Assuming the respondents were chosen as follows:

Scenario 1
M M M M M F F F F F

Scenario 2
F M F M F M F M F M

Scenario 3
F F F M M F M M F F

Scenario 1 has only 2 runs and therefore the scenario cannot be considered random because there are too few runs.

Scenario 2 has too many runs, 5 runs. And therefore would not be considered as random

Scenario 3 has 5 runs and therefore we need to perform a test to determine the randomness of the data.

Run Test Procedure

First we need to assume that the data available for the analysis consists of a sequence of observations, recorded in order of occurrence, which we can categorize into two mutually exclusive types.

First, you need to determine the total sample size, then the number of observation of each type as presented below:

n = total sample size
n_1 = the number of observation of one type
n_2 = the number of observations of the other type

Hypothesis

Then State the null and alternate hypothesis

A. TWO-SIDED
   H_0: the pattern of occurrence is random
   H_1: the pattern of occurrence is not random
B. B. ONE-SIDED
   H_0: the pattern of occurrence is random
   H_1: the pattern of occurrence is not random (because there are too few runs to be attributed as random)

Test Statistic and Decision Rule

The test statistic is r = total number of runs

The decision rules is also called the acceptance or rejection criteria. It depends on the test statistic (calculated) and the values of upper and lower limits (from statistical tables)

A. TWO-SIDED	B. ONE-SIDED (H_1 has few runs)	C. ONE-SIDED (H_1 has to many runs)
Reject H_0:	**Reject H_0:**	**Reject H_0:**
$R \leq$ lower critical value OR $R \geq$ upper critical value	$R \leq$ lower critical value	$R \geq$ upper critical value

Critical Value

Critical value is determined from statistical table using n_1 and n_2

Example 1

On a commuter train, the conductor wants to see whether the passengers entering a train enter in a random manner. He observes the first 25 people, with the following sequence of males (M) and females (F).

F F F M M F F F F M F M M M F F F F M M F F F M M

Test for randomness at $\alpha = 0.05$

Solution Steps

Step 1: State the null and alternate hypothesis

H_0: The patter of occurrence of males and females enter the train is random

H_1: The pattern of occurrence of males and females entering the train is not random

Step 2: Find the test statistic (number of runs)

You can easily get this by grouping each run as shown below:

FFF MM FFFF M F MMM FFFF MM FFF MM

Test statistic, r = 10

n_1 = number of females = 15

n_2 = number of males = 10

Step 3: Find the critical value

we can find the lower and upper critical value from statistical run table

$n_1 = 15$, $n_2 = 10$

Lower critical value = 7

Upper critical value = 18

Step 4: Make your decision

since r = 10 which is between 7 and 18, we accept the null hypothesis (we fail to reject it)

Step 5: Draw a Conclusion

There are not enough evidence to reject the claim that the pattern of occurrence of males and females enter the train is determined by a random process.

Exercise

We have 20 people that enrolled in a drug abuse program. Test the claim that the ages of the people, according to the order in which they enroll occur at random, at $\alpha = 0.05$.

The data are as follows:
18, 36, 19, 22, 25, 44, 23, 27, 27, 35, 19, 43, 37, 32, 28, 43, 46, 19, 20, 22

Exercise

Table below shows the departures from normal of daily temperatures recorded at Atlanta, Georgia during February 1969. We would like to know whether we may conclude that the pattern of departures above and below normal is the result of a non-random process.

Day	Departure from normal	Day	Departure from normal
1	12	16	-9
2	13	17	6
3	12	18	7
4	11	19	10
5	5	20	10
6	2	21	1
7	-1	22	1
8	2	23	3
9	-1	24	7
10	3	25	-2
11	2	26	-6
12	-6	27	-6
13	-7	28	-5
14	-7	29	-2
15	-12	30	-2

Solution Steps

Step 1: State the null and the alternate hypothesis

H_0: The pattern of occurrence of negative and positive deviations from normal is determined by a random process

H_1: The pattern of occurrences of negative and positive deviations from normal is not determined by a random process (claim)

Step 2: Find the test statistics (number of runs)

To get the number of runs, we need to find the departures from normal above and below zero. The departures from normal that is above 0 are recorded as A and those that are below 0 are recorded as B.

If we do this we would have the arrangement as follows:

AAAAAA B A B AA BBBBB AAAAAAAA BBBBBB

Test Statistic (number of runs) r = 8

n_1 = number of A = 17

n_2 = number of B = 13

Step 3: Find the critical value

$n_1 = 17$, $n_2 = 13$

Using statistical table we find the:

Lower critical value = 10

Upper critical value = 22

Step 4: Make the decision

since r = 8, which is lower than the critical value, we reject the null hypothesis (H_0)

Step 5: Draw the conclusion

there is enough evidence to support the claim that the pattern of occurrence of positive and negative departures from normal is not random

Mann Whitney U Test (Wilcoxon Rank Sum Test)

A popular nonparametric test to compare outcomes between two independent groups is the Mann Whitney U test. The Mann Whitney U test, sometimes called the Mann Whitney Wilcoxon Test or the Wilcoxon Rank Sum Test, is used to test whether two samples are likely to derive from the same population (i.e., that the two populations have the same shape). Some investigators

interpret this test as comparing the medians between the two populations. Recall that the parametric test compares the means (H_0: $\mu_1=\mu_2$) between independent groups.

In contrast, the null and two-sided research hypotheses for the *nonparametric test* are stated as follows:

H_0: The two populations are equal versus

H_1: The two populations are not equal.

This test is often performed as a two-sided test and, thus, the research hypothesis indicates that the populations are not equal as opposed to specifying directionality. A one-sided research hypothesis is used if interest lies in detecting a positive or negative shift in one population as compared to the other. The procedure for the test involves pooling the observations from the two samples into one combined sample, keeping track of which sample each observation comes from, and then ranking lowest to highest from 1 to n_1+n_2, respectively.

Test Statistic for the Mann Whitney U Test

The test statistic for the Mann Whitney U Test is denoted **U** and is the ***smaller*** of U_1 and U_2, defined below.

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

where R_1 = sum of the ranks for group 1 and R_2 = sum of the ranks for group 2.

Example:

A new approach to prenatal care is proposed for pregnant women living in a rural community. The new program involves in-home visits during the course of pregnancy in addition to the usual

or regularly scheduled visits. A pilot randomized trial with 15 pregnant women is designed to evaluate whether women who participate in the program deliver healthier babies than women receiving usual care. The outcome is the APGAR score, measured 5 minutes after birth. Recall that APGAR scores range from 0 to 10 with scores of 7 or higher considered normal (healthy), 4-6 low and 0-3 critically low. The data are shown below.

Usual Care	8	7	6	2	5	8	7	3
New Program	9	9	7	8	10	9	6	

Is there statistical evidence of a difference in APGAR scores in women receiving the new and enhanced versus usual prenatal care? We run the test using the five-step approach.

- **Step 1.** Set up hypotheses and determine level of significance.

  H_0: The two populations are equal versus

  H_1: The two populations are not equal. $\alpha = 0.05$

- **Step 2.** Select the appropriate test statistic.

Because APGAR scores are not normally distributed and the samples are small (n_1=8 and n_2=7), we use the Mann Whitney U test. The test statistic is U, the smaller of

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \text{ and } U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

Where R_1 and R_2 are the sums of the ranks in groups 1 and 2, respectively.

- **Step 3.** Set up decision rule.

The appropriate critical value can be found in the table above. To determine the appropriate critical value we need sample sizes (n_1=8 and n_2=7) and our two-sided level of significance (α=0.05). The critical value for this test with n_1=8, n_2=7 and $\alpha = 0.05$ is 10 and the decision rule is as follows:

Reject H_0 if $U \leq 10$.

- **Step 4.** Compute the test statistic.

The first step is to assign ranks of 1 through 15 to the smallest through largest values in the total sample, as follows:

		Total Sample (Ordered Smallest to Largest)		Ranks	
Usual Care	New Program	Usual Care	New Program	Usual Care	New Program
8	9	2		1	
7	8	3		2	
6	7	5		3	
2	8	6	6	4.5	4.5
5	10	7	7	7	7
8	9	7		7	
7	6	8	8	10.5	10.5
3		8	8	10.5	10.5
			9		13.5
			9		13.5
			10		15
				R_1=45.5	R_2=74.5

Next, we sum the ranks in each group. In the usual care group, the sum of the ranks is R_1=45.5 and in the new program group, the sum of the ranks is R_2=74.5. Recall that the sum of the ranks will always equal n(n+1)/2. As a check on our assignment of ranks, we have n(n+1)/2 = 15(16)/2=120 which is equal to 45.5+74.5 = 120.

We now compute U_1 and U_2, as follows:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 8(7) + \frac{8(9)}{2} - 45.5 = 46.5$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 8(7) + \frac{7(8)}{2} - 74.5 = 9.5$$

Thus, the test statistic is U=9.5.

- **Step 5.** Conclusion:

We reject H_0 because $9.5 \leq 10$. We have statistically significant evidence at $\alpha = 0.05$ to show that the populations of APGAR scores are not equal in women receiving usual prenatal care as compared to the new program of prenatal care.